Complete state and fidelity during two exchanges: $\tau = 100/\mathrm{meV}$, $t_c = E_0 = 0.3$ meV, $K = 2A$

Ideal reference: $E_1 = |t_1| = 0$

$E_1 = 0.01, \quad |t_1|/E_1 = 0.1$

$E_1 = 0.01, \quad |t_1|/E_1 = 1$

$E_1 = 0.01, \quad |t_1|/E_1 = 10$

Full state $\mathcal{B}$

$d = B_{12}$

$x_1 = B_{13}$

$x_2 = B_{1a}$

$x_3 = B_{1b}$

$y_1 = B_{23}$

$y_2 = B_{2a}$

$y_3 = B_{2b}$

$h_1 = B_{3a}$

$h_2 = B_{3b}$

$h_3 = B_{ab}$

B_{ij}

1.0

0.5

0.0

−0.5

−1.0

Squared fidelity F

$|\langle\psi_1^+(0)|\psi_1^-(t)\rangle|^2$

$|\langle\psi_1^-(0)|\psi_1^-(t)\rangle|^2$

Overlap modulus $f = \sqrt{F}$

$|\langle\psi_1^+(0)|\psi_1^-(t)\rangle|$

$|\langle\psi_1^-(0)|\psi_1^-(t)\rangle|$

0.0

0.5

1.0

0 1 2 3 4 5 6

step (t/τ)